

APPROACH AND MANAGEMENT OF JOINT PAIN IN PRIMARY CARE

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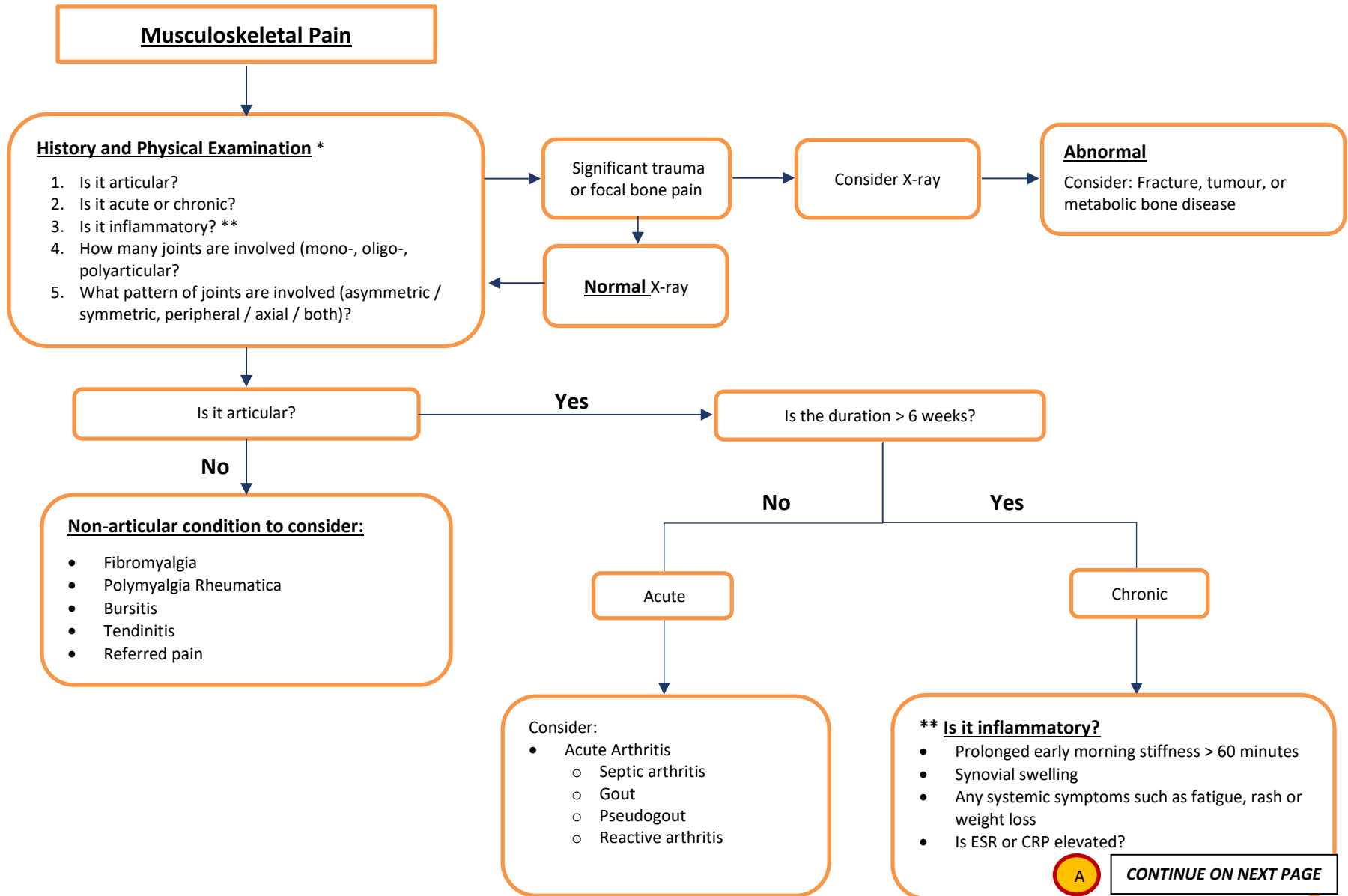
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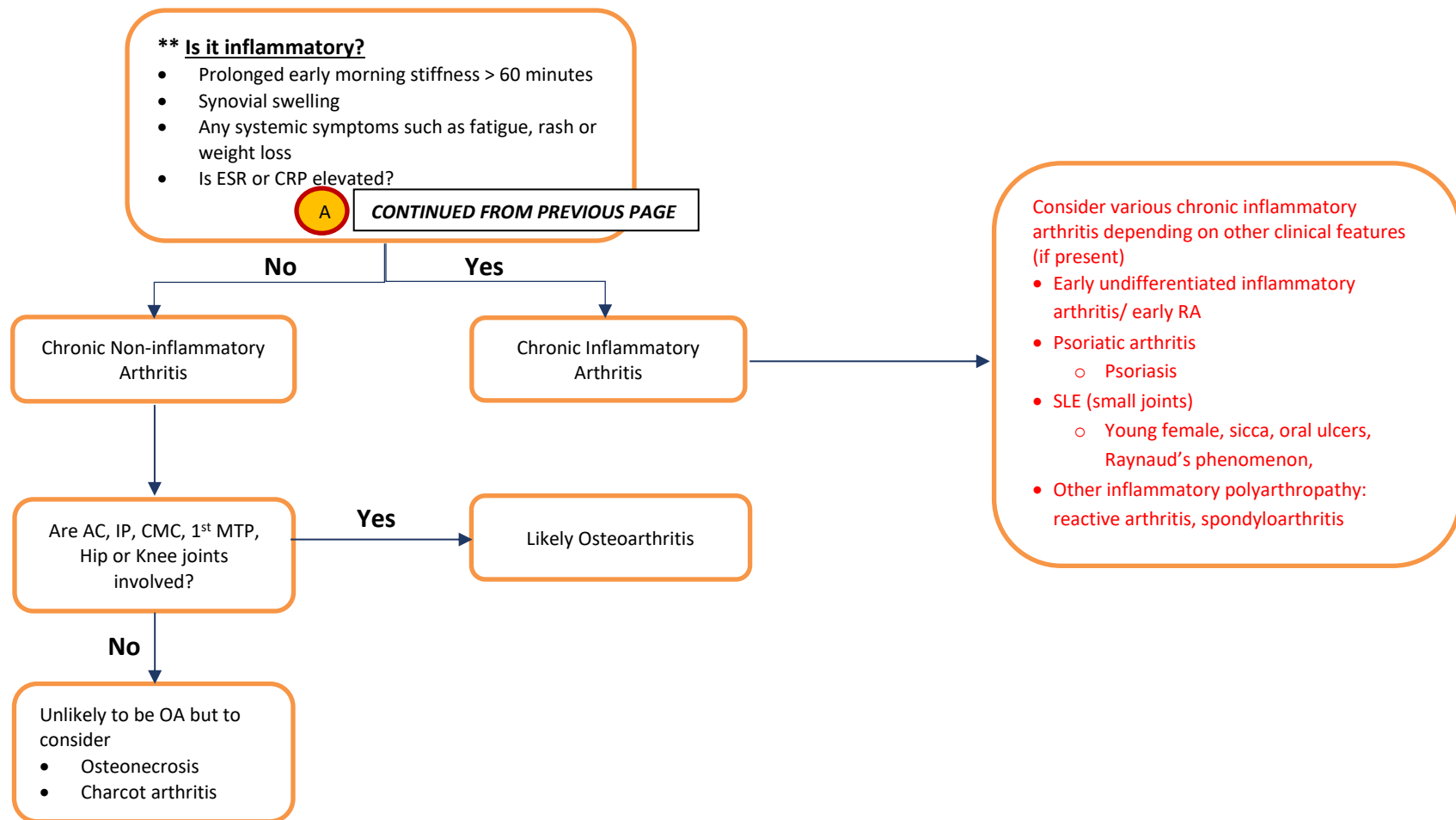
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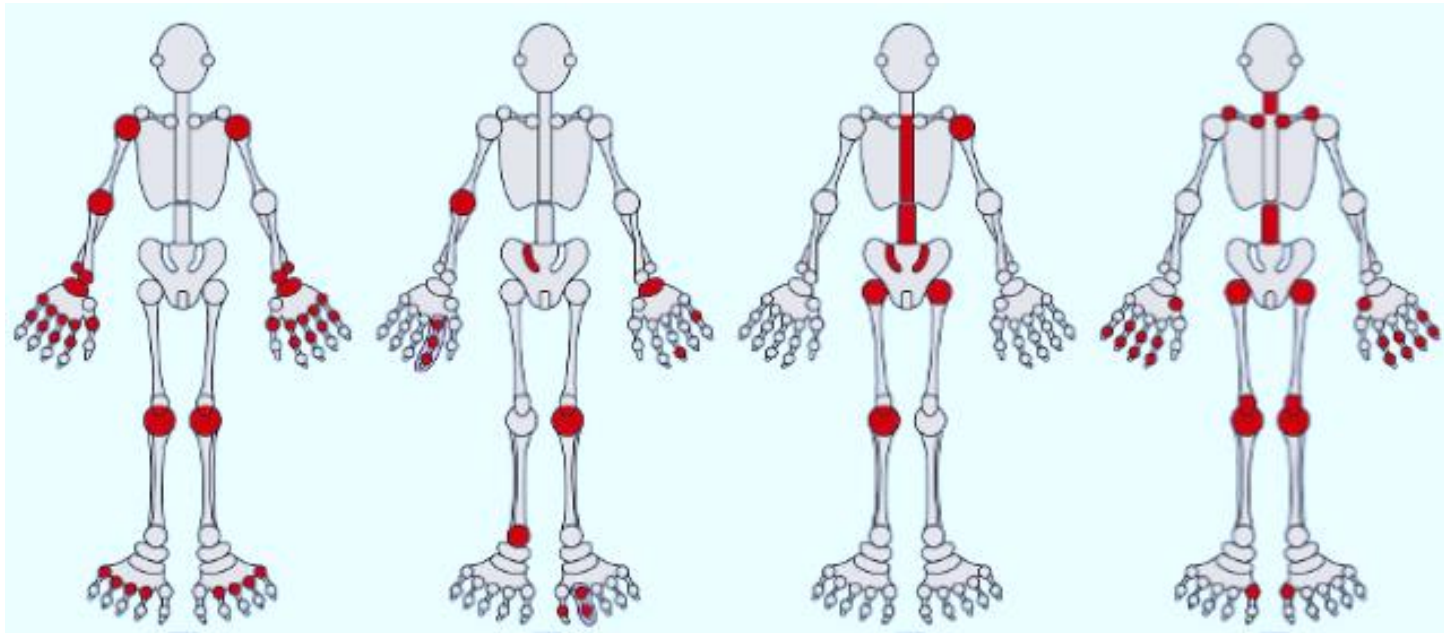
WORKFLOW FOR APPROACH & MANAGEMENT OF MUSCULOSKELETAL PAIN





HISTORY

1. The pattern of joints involved, i.e., small and/or large joints, asymmetrical or symmetrical
2. Duration of pain and any associated symptoms of stiffness, numbness or weakness
 - Early morning stiffness lasting more than 1 hour is significant
3. Any aggravating and relieving factors of the painful joint and its progression of the symptoms
 - Classically inflammatory joint pain improves with activity, while non-inflammatory joint pain worsens with activity
4. Any axial involvement (spine, sacroiliac joints)
5. Family history of inflammatory arthritis, connective tissue disease, skin psoriasis
6. Other constitutional/systemic symptoms (i.e., fever, fatigue, chills, body ache, nausea, weight loss, diarrhoea, skin rashes, mucosal ulcers, uveitis, *etc.*)



Rheumatoid arthritis (RA)

Psoriatic arthritis (PsA)

Spondyloarthritis (SpA)

Osteoarthritis (OA)

From: <https://dundeemedstudentnotes.wordpress.com/2014/06/16/polyarthritis/>

EXAMINATION OF AFFECTED JOINTS

1. **LOOK**: Inspect the joint for physical deformity, site(s) of swelling, **skin changes suggestive of infection**, any past surgical scars, pattern of joint involvement
2. **FEEL**: Palpate the joint and assess for increased skin temperature, joint tenderness, joint swelling and joint deformity
3. **MOVE**: Assess the joint for active and passive range of movement, power and weakness. One should not forget to assess the functional status of the affected **regions (see below)**
4. **SPECIAL TESTS**: “Metacarpophalangeal (MCP) or metatarsophalangeal (MTP) squeeze test”, also known as the Gaenslen’s test (GT)²: Easy and simple to perform routinely in a busy clinical setting, involves applying pressure on MCP / MTP heads region, like in a hand-shake and assessing for tenderness. A positive squeeze test signifies the presence of synovitis at the MCPJs / MTPJs
5. **ASSESS FUNCTION**: Assess the functional status of the hands/lower limbs: (5 simple tasks to screen hand function: Turn the doorknob, use a key, or buttoning, pick up a coin, and write using a pen. Lower limbs: Assess the gait/check for falls, get up and go test)

Other Related Physical Examination

Hallmarks features suggestive of synovitis

- Synovial swelling
- Warmth/tenderness over a joint
- Joint effusion
- Limitation in range of motion

Signs of Degenerative Joint Disease	Signs of Inflammatory Joint Disease
• Bony overgrowth of joints (osteophytes) • Limited range of movement • Crepitus during action / passive motion (usually knees, although crepitus does not necessarily mean degeneration)	• Bone erosions • Limited range of movement • Erythema and warmth, e.g., gout, pseudogout, infection • Joint effusion • Joint tenderness
• Joint deformity – e.g., Varus deformity consistent with knee osteoarthritis (look for restriction of movement, mal-alignment of articulating bones, alteration in relation to the 2 articulating surfaces)	• Joint deformity is only seen in chronic uncontrolled inflammatory arthritis
• Pain more mechanical in nature especially with activities such as squatting, climbing stairs etc.	• Pain not relieved significantly by rest
 Picture 1. Features of Osteoarthritis. American Family Physician, 2012;85(1):49-56	 Figure 1. Autoimmune Diseases Volume 2013 (2013), Article ID 185340, 6 pages http://dx.doi.org/10.1155/2013/185340

Additional Clinical Features to Take Note

1. The presence of subcutaneous nodules over the elbows may be due to rheumatoid nodules or tophi (*Picture 2*).
2. Skin lesions may suggest that the joint symptoms are due to psoriatic arthritis, SLE, viral infection, or Adult Onset Still's disease/ systemic Juvenile Idiopathic Arthritis.
3. Eye involvement - including keratoconjunctivitis sicca, uveitis, conjunctivitis, and episcleritis, are features suggestive of rheumatic illnesses.
4. Concomitant axial pain or stiffness suggests the possibility of axial spondyloarthritis or another seronegative spondyloarthritis.
5. Causes of inflammatory arthritis ([Table 1](#)) – many different causes to consider.
6. Check other relevant organ systems for possible conditions related to rheumatoid condition: e.g., Interstitial fibrosis, Felty's Syndrome, *etc.*



Picture 2. Subcutaneous nodules over the elbow. Reprinted from the Clinical Slide Collection on the Rheumatic Diseases.

Infectious arthritis	Crystal-Induced Arthritis
Bacterial	Juvenile Idiopathic Arthritis
Lyme Disease	Systemic Rheumatic Illness
Bacterial Endocarditis	Systemic Lupus Erythematosus
Viral	Systemic Vasculitis
Other infection	Systemic Sclerosis
	Polymyositis/Dermatomyositis
Post-infectious (reactive) arthritis	Adult Onset Still's Disease
Rheumatic Fever	Behcet's Disease
Reactive Fever	Relapsing Polychondritis
Enteric Infection	Autoinflammatory disorder
Other seronegative spondyloarthritides	Other Systemic illness
Ankylosing Spondylitis	Sarcoidosis
Psoriatic Arthritis	Palindromic Rheumatism
Inflammatory bowel disease	Malignancy
Rheumatoid arthritis	Hyperlipoproteinemias
Inflammatory osteoarthritis	Familial Mediterranean Fever

Table 1. Causes of polyarticular inflammatory arthritis⁷

RISK FACTORS FOR SEPTIC ARTHRITIS

Always consider this in the acute presentation of inflammatory monoarthritis:

- Elderly and very young children
- Skin infection and cutaneous ulcers
- Diabetes mellitus (especially if poorly controlled)
- Chronic kidney disease or liver disease
- Previous joint pathology (e.g., rheumatoid arthritis, osteoarthritis, crystal arthropathy)
- Recent joint surgery
- Prosthetic joint
- IV drug abuse, alcoholism
- Previous intra-articular corticosteroid injection
- Recent hospitalisation with risk for haematogenous spread of infection, e.g., intravenous cannulation, PICC lines

INVESTIGATIONS

1. Blood Investigations

Full Blood Count (FBC)

- High total white cell count and differential count: acute gout/ septic arthritis
- Anaemia – normochromic, normocytic (chronic illness), microcytic (consider iron deficiency from NSAID use)
- Thrombocytosis – either reactive or underlying iron deficiency
- Anaemia, leukopenia, lymphopenia, thrombocytopenia - consider acute viral infection, SLE or adverse effects of DMARD in-patient with established inflammatory arthritis on DMARD.

Erythrocyte Sedimentation Rate (ESR) and/or C-reactive protein (CRP)

- Help to differentiate presence of inflammatory medical conditions. However, a normal ESR / CRP may not exclude inflammatory arthritis.
- There are rare instances where patients with inflammatory arthritis may have a normal ESR and/or CRP, even in the presence of active peripheral arthritis or spondylitis.

Note: Upper limit of normal of ESR: Male = Age / 2; Female = (Age+10) / 2

Rheumatoid Factor (RF) should not be ordered routinely but should be reserved for cases in which there is a reasonable clinical suspicion, i.e., when there are prolonged inflammatory joint pains (acute joint pains should be first evaluated to exclude infective or crystal disease causes) with / without systemic symptoms. The indiscriminate use of RF will result in a high frequency of false-positive results and in additional expensive and unnecessary testing.

Antinuclear Antibody (ANA) - not available in NUP should not be ordered routinely but should be reserved for cases in which there is a reasonable clinical suspicion for a connective tissue disease, i.e., when there are systemic symptoms and/or laboratory features (e.g., cytopenias) **IN ADDITION** to the arthritis.

2. Radiological Imaging

Consider radiographs when:

1. Significant history of trauma (exclude fracture / dislocation)
2. Focal bone pain (to exclude fracture or neoplasm)

Plain X-Ray - able to detect fractures, tumours, and metabolic bone disease.

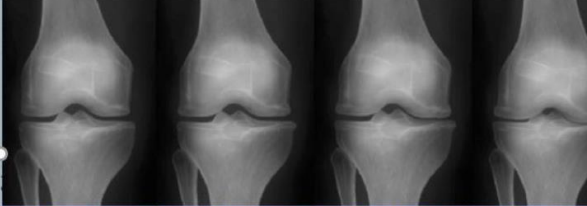
A plain X-ray is unnecessary to diagnose osteoarthritis in patients with risk factors and typical symptoms and signs. It can be considered in patients with chronic atraumatic joint pain (> 6 weeks) and may be helpful before referral for surgery as a baseline. It is essential to appreciate that radiological findings may not correlate well with patients' symptoms

When ordering X-rays to assess knee osteoarthritis:

1. A bilateral weight-bearing anterior-posterior (AP) view and lateral (LAT) view can be considered to accentuate the radiological features of osteoarthritis and for comparison
2. Skyline view should be considered if involvement of the patellofemoral compartment is suspected

The Kellgren-Lawrence grading system

- Grade 0 (none): no X-ray changes of OA
- Grade 1 (doubtful): doubtful joint space narrowing and possible osteophytic lipping
- Grade 2 (minimal): definite osteophytes and possible joint space narrowing
- Grade 3 (moderate): moderate multiple osteophytes, definite narrowing of joint space, some sclerosis and possible deformity of bone ends
- Grade 4 (severe): large osteophytes, marked narrowing of joint space, severe sclerosis and definite deformity of bone ends



	Grade 1	Grade 2	Grade 3	Grade 4	
Classification	Normal	Doubtful	Mild	Moderate	Severe
Description	No Features of OA	Minute Osteophyte: doubtful significance	Definite Osteophyte: normal joint space	Moderate Joint Space Reduction	Joint Space Greatly Reduced: subchondral sclerosis

Hand X-Rays



Rheumatoid Arthritis

- PIP, MCP, radial styloid
- Periarticular osteopenia (synovial hyperplasia)
- Joint space widening
- Marginal erosions
- Boutonniere, swan neck, wrist and MCP deformities



Osteoarthritis

- DIP, PIP, 1st CMC
- Osteosclerosis
- Heberden's and Bouchard's nodes
- DIP and PIP involvement



Gout

- Tophi
- Giant erosions with overhanging margins



Pseudogout

- Chondrocalcinosis

<http://www.medinterestgroup.com/wp-content/uploads/2012/12/Slide013.jpg>

Do note that changes on x-ray in patients with inflammatory arthritis and crystal arthritis are only apparent in chronic disease (usually years).

The following investigations are only done in the hospitals or tertiary institutions (Not available at NUP at this time):

1. **Musculoskeletal Ultrasound (MSUS)**: To assess soft tissues, cartilage, bone surfaces, and fluid-containing structures. MSUS can be used clinically for:
 - Assessing for the presence of synovitis (e.g. where ESR/CRP is normal and patient not keen to escalate treatment)
 - Imaging tendons and bursae to guide treatment e.g. supraspinatus tendinopathies, rotator cuff tears, sub-acromial / sub-deltoid bursitis
 - Guiding aspiration and/or injection of joints or soft tissues, e.g. in rheumatoid joints with severe damage, Achilles tendinopathies, plantar fasciitis. *

** Blind joint and soft tissue aspirations/injections are still advocated in straightforward cases*
2. **Computerised Tomography (CT)**: May be useful in detecting cortical bony lesions and is widely used in fracture detection, especially in complex areas such as the cervical spine.
3. **Magnetic Resonance Imaging (MRI)**: Commonly used in the following situations in a rheumatology/orthopaedic clinic:
 - MRI sacroiliac joints - diagnosis of early spondyloarthritis especially when ESR/CRP normal
 - MRI knee - for internal derangement of the knee, ligamentous injuries, meniscal tears following sports injuries or in osteoarthritis of the knees
 - MRI spine - for mechanical back pain with increasing severity, sciatica, radicular symptoms; osteoporotic compression fractures (acute fractures which may be amenable to vertebroplasty), or when there are red flags for malignancy
 - MRI shoulder - for rotator cuff tears prior to surgery
4. **Nuclear Medicine Studies**: Bone Scan for metastases, cancer staging especially for osteoblastic tumours like CA breast and prostate
5. **Synovial Fluid Composition**
 - Inflammatory joint fluid with crystals establishes the diagnosis of gout or pseudogout.
 - Non-inflammatory synovial fluid (e.g. < 2000 WBCs or < 75 percent neutrophils) should lead to consideration of osteoarthritis, soft tissue injury, or viral infection.
 - A positive synovial fluid culture establishes the diagnosis of infectious arthritis.
 - A sterile inflammatory joint fluid raises the suspicion of systemic rheumatic disorders; such patients should have further evaluation and should be referred to rheumatologist for further workup.
 - A bloody effusion should lead to consideration of a trauma (most commonly meniscal injuries in OA knee), coagulopathy, tumour, or a Charcot joint; subsequent evaluation includes a prothrombin (PT), partial tissue thromboplastin time (PTT), platelet count, and bleeding time.

MUSCULOSKELETAL EMERGENCIES

Red flags:

1. Hot or swollen joints may suggest infection; always consider TB (extrapulmonary) in a patient with chronic arthritis and risk factors, e.g. on an anti-TNF biologic, regardless of duration of biologic use.
2. Constitutional symptoms (high-grade fever, weight loss, malaise) suspicion of infection or sepsis; in children, acute leukaemia is always a consideration.
3. Weakness may be a symptom of a compartment syndrome or an acute myelopathy.
4. Burning pain, numbness, or paraesthesia may suggest an acute myelopathy, radiculopathy, or neuropathy.

MANAGEMENT – WITH A FOCUS ON OA

Osteoarthritis

Initial management

- Conservative management should be the first line and mainstay management as most patients with OA can be managed in the community

Treatment goals

- Adequate pain control
- Improve or maintain function
- Allow patients to function independently in the community for as much as possible

Non-pharmacological Treatment

- Patient education
- Lifestyle modification: appropriate weight loss, low-impact exercises (e.g., swimming, cycling (including stationary bikes), deep-water running – aqua jog)
- Physiotherapy: Pain education, activity modification, lower limb strengthening exercises including quadriceps strengthening
- Appropriate walking aids

Pharmacological Treatment

- Analgesia:
 - Topical NSAIDs
 - Oral paracetamol
 - Oral NSAIDs (if no contraindication)
 - Oral opioids have similar pain reduction efficacy to NSAIDs but with a relatively higher incidence of side effects. Should not be given as first line.
- Intraarticular injections (Hydrocortisone & Lignocaine, viscosupplementations – Hyaluronic acid) may provide temporary pain reduction and relief but are not the mainstay of conservative treatment. Their efficacy and cost effectiveness remain **controversial**.
- The evidence surrounding the value of nutritional supplements, including glucosamine and/or chondroitin, in managing patients with osteoarthritis, has been **inconclusive**. The weight of the evidence indicates a lack of efficacy and essentially placebo effects

Roles of non-doctor team members

- Physiotherapists play a crucial role in the management of OA
- They can provide patient education such as pain education and promote self-management
- They can provide directed muscle strengthening exercises to reduce pain and improve function
- They can provide advice on activity modification and assisted devices (e.g. walking aid)

Surgical Options

- Realignment osteotomy
- Arthroplasty

Post Surgical Care

- Pain management
- Wound care
- Rehabilitation: usually initiated postoperatively inpatient. It includes range of motion exercises, gait training, quadriceps strengthening and training in activities of daily living.
- In the long term: pursue low- to moderate- intensity, low-impact exercises such as walking and swimming.

Consideration for referral to hospital and a specialist

Most patients with osteoarthritis can be managed in primary care.

Referral to a specialist should be discussed with patients with unsatisfactory improvement of pain, stability, or function despite adequate conservative (non-pharmacological and pharmacological) treatment.

RECOMMENDED CARE COMPONENT FOR OSTEOARTHRITIS

Recommended Care Components	Minimum Frequency*	Remarks
Assessment of joint pain	Annually	Visual Analog Scale
Weight and BMI Assessment (less relevant for hand OA)	Annually	Keep < 23kg/m ² (For non asian population, keep BMI < 25kg/m ²)
Activities of Daily Living (ADL) Assessment (if appropriate)	Annually	Helps to determine functional impairment Referral to physiotherapy or occupational therapy assessment for assisted devices should be considered if function is impaired

** More frequently if clinically indicated*

MANAGEMENT – WITH A FOCUS ON RA

Rheumatoid Arthritis

Non-pharmacological Treatment

- Patient education
- Psychosocial interventions
- Rest, exercise, and physical and occupational therapy
- Nutritional and dietary counselling
- Interventions to reduce risks of cardiovascular disease, including smoking cessation and lipid control
- Screening for and treatment of osteoporosis
- Immunizations to decrease risk of infectious complications of immunosuppressive therapies

Pharmacological Treatment

The aims of using pharmacological/immunosuppressive therapy in RA are:

- To induce or maintain a remission
- To reduce the frequency of flare or relapse
- To allow tapering of glucocorticoids while maintaining disease control
- Prevention of progressive joint erosions, damage and loss of function

1. Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) / Selective Cyclooxygenase Inhibitors

- For pain relief in RA, but do not prevent irreversible joint damage
- Caution is advised as long term and large dosages of NSAIDs will affect renal function and increase the risk of peptic ulcer disease / upper gastrointestinal bleeding and cardiovascular events
- Baseline Hb and Cr should be checked
- Patients with cross-reactive NSAID hypersensitivity (angioedema/urticaria/anaphylaxis to 2 or more NSAIDs) should be treated with paracetamol (if tolerated) or opioids / tramadol for pain relief
- Referral to an allergist for drug provocation test to a selective COX-2 inhibitor to assess for tolerance is recommended should a selective COX-2 inhibitor be needed

2. Glucocorticoids

- Effective in suppressing the symptoms of RA and have an impact on disease progression
- However, because of their associated toxicities, they are not ideal for the long-term management of RA
- A trial of Prednisolone 5 mg BD for 2 weeks can be used to assess response in newly diagnosed RA

3. Non Biologic Disease-Modifying Anti-Rheumatic Drugs (DMARDs)

- Treatment with DMARDs should be initiated as soon as possible for patients with confirmed RA and persistent synovitis; as active RA may lead to irreversible joint damage early in the disease process.
- Combination DMARD therapy is sometimes necessary to achieve disease control in patients with severe RA and may also be appropriate for some patients with early, moderately active disease.
- Such patients should be referred to a rheumatologist for early management.
- Commonly used non-biologic DMARDs that are used in Singapore are:
 - Hydroxychloroquine, sulfasalazine, methotrexate – alone or in combination
 - Leflunomide, cyclosporine – when the above first line DMARDs fail or patients develop adverse effects
 - Intramuscular gold - when the above first line DMARDs fail or patients develop adverse effects.

4. Biologic DMARDs and targeted synthetic DMARDs

- Generally, targets cytokines, their receptors or other cell surface molecules that mediate the inflammatory response and bone resorption. (e.g. tumour necrosis factor (TNF)-alpha)
- The most commonly used biologic agents in Singapore are:
 - Anti-TNF inhibitors - infliximab (IV), adalimumab (SC self-injection every 2 weeks), golimumab (SC self-injection once a month)
 - Rituximab – IV, usually 2 doses 2 weeks apart for refractory RA
 - Others –tocilizumab (anti-IL6)
- The most commonly used targeted synthetic DMARDs in RA is tofacitinib and baricitinib

Common side effects of all the above:

- Infections – including bacterial infections, extrapulmonary TB, herpes zoster
- Malignancy

Consideration for collaborative care with specialist

- Collaborative care or anchoring care with primary care physicians can be considered for:
 - Patients deemed to be in DMARD-free remission
 - Patients deemed to have quiescent/low disease activity (no swollen and/or tender joints, ESR/CRP within normal range) for at least 3 – 6 months under a specialist's care
 - Patients on non-biologic DMARD therapy at a maintenance dosage
- Patients on active treatment for RA should be on follow up with a rheumatologist.
 - a. Management of any concurrent osteoporosis, and other appropriate preventive care (e.g., vaccinations, cardiovascular risk factors) can continue at primary care
 - b. Please refer to the appropriate NUP CPGs:
 - [Hypertension](#)
 - [Lipids](#)
 - [Diabetes Mellitus](#)
 - [Chronic Kidney Disease](#)
 - [Osteoporosis](#)
- Patients in the following clinical scenarios should also be arranged for active review by a rheumatologist
 - Patients requiring new initiation of DMARDs
 - Patients with RA flares requiring either high dose (e.g., prednisolone >10mg/day) or long term (≥6 months) glucocorticoid therapy (which should be accompanied by appropriate dose adjustment of DMARDs)
 - Patients with extra-articular manifestations of RA
 - Patients on biologic DMARD therapy
 - Paediatric patients with 6 weeks or more of persistent joint swelling, and joint pain
 - Patients who develop active disease (1 or more swollen and/or tender joints, high ESR/CRP) while on collaborative care

Monitoring required and considerations for patients getting DMARDs from Rheumatologist

- Commonly all RA patients prescribed with immunosuppressive agents have the following monitored:
 - FBC, ESR, AST and ALT, and Creatinine done at 2-3 monthly intervals when stable;
 - FBC, ESR, AST and ALT, and Creatinine within 2-4 weeks of dose escalation.
 - Abnormal FBC, increasing ESR and increased transaminitis (ALT /AST) 3x higher than normal require further evaluation and assessment.
 - Eye assessment and evaluation needs to be done for patients on chronic hydroxychloroquine use (Retinal toxicity)
- Trend of the laboratory result is as important as the absolute value. The following situations may necessitate reduction of DMARD dosage, e.g.:
 - Gradual decline in TW: for e.g. $4.4 \times 10^6/L$ to $3.6 \times 10^6/L$
 - ALT increasing gradually: for e.g. 30 U/L to 56 U/L to 78 U/L when on methotrexate
 - Gradual rise in Cr from baseline to 1.5-2x upper limit of normal when on cyclosporine/methotrexate.
 - Medication review of all other medications the patient is taking for his/her chronic medical conditions may potentiate these laboratory abnormalities. DMARD interruption/dose alteration may be needed, e.g.
 - Patient on MTX and statin for hyperlipidaemia with underlying non-alcoholic fatty liver disease
 - Gradual rise in Cr from ACE inhibitor / ARB dose escalation when on cyclosporine/methotrexate
- An acute viral illness may cause transient leukopenia, elevated ALT / AST which may be aggravated in patients on pre-existing DMARDs.
- This may require temporary interruption of the DMARD for 1 week till leukopenia and/or elevated ALT / AST resolve.

RECOMMENDED CARE COMPONENT FOR RHEUMATOID ARTHRITIS

Recommended Care Component	Minimum Frequency*	Remarks
Assessment of RA Disease Activity	Annually	Number of tender / swollen joints, CRP or ESR; Measures of disease activity must be obtained and documented regularly, as frequently as monthly for patients with high/moderate disease activity, or less frequently (at least at 6 - 12 months intervals) for patients in sustained low disease activity or remission.

** More frequently if clinically indicated*

SUMMARY

1. Patients with a history of significant trauma or focal bone pain should have plain radiographs of the affected joint to rule out fracture, tumour, or metabolic bone disease.
2. Good history and relevant targeted physical examination are important when approaching patient presenting with acute or chronic joint pain.
3. Most basic investigations including blood tests can be non-specific and need to be considered in tandem with patient's history and physical examination findings.
4. Patients need to be referred urgently to the rheumatologist or to the Emergency Department (if severe / very unwell) for:
 - Suspected active SLE / Connective tissue disease (CTD)
 - Active dermatomyositis complicated by interstitial lung disease (ILD)
 - Known antiphospholipid syndrome with DVT / PE or stroke

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USEFUL REFERENCES AND READING MATERIALS

1. Exercise Sheets and Videos: <https://www.versusarthritis.org/about-arthritis/exercising-with-arthritis/>
2. Clinical Assessment of the Musculoskeletal System: <https://www.versusarthritis.org/about-arthritis/healthcare-professionals/training-and-education-resources/clinical-assessment-of-patients-with-musculoskeletal-conditions/>
3. Drug Information: <https://www.versusarthritis.org/about-arthritis/treatments/drugs/>